

Lab Setup Report — Task 1

Foundation & Environment Setup

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Task:	Task 1 — Foundation & Environment Setup		Days 1–12
Date:	2026-06-07	Status:	Completed ✓

1. Objective

The objective of Task 1 is to build strong fundamentals in cybersecurity, networking, cryptography, and set up a professional penetration testing lab environment using VirtualBox, Kali Linux, and Metasploitable2. This forms the foundation for all subsequent tasks in the internship.

2. Lab Environment Setup

Component	Details	Status
Host Machine	Windows PC — VirtualBox 7.x	✓ Installed
Attacker Machine	Kali Linux (VirtualBox Image)	✓ Running
Target Machine	Metasploitable2 (.vmdk)	✓ Running
Network Type	Host-Only Adapter (Private Lab Network)	✓ Configured
Kali IP Address	192.168.56.xxx	✓ Active
Metasploitable2 IP	192.168.56.xxx	✓ Active

3. Cybersecurity Basics Learned

3.1 CIA Triad

Principle	Description
Confidentiality	Ensuring only authorized users can access data
Integrity	Ensuring data is not tampered with or altered
Availability	Ensuring systems and data are accessible when needed

3.2 Threat Types Explored

- Phishing — Fake emails/websites to steal credentials
- Malware — Malicious software (viruses, trojans, spyware)
- DDoS — Distributed Denial of Service attacks to crash servers
- SQL Injection — Injecting malicious SQL into database queries
- Brute Force — Trying all possible passwords until one works
- Ransomware — Encrypts victim files and demands payment

3.3 Attack Vectors Studied

- Social Engineering — Manipulating people to reveal confidential info
- Wireless Attacks — Targeting Wi-Fi networks (Evil Twin, Deauth)
- Insider Threats — Attacks from within an organization

4. Linux Fundamentals Practiced

Category	Commands Used	Purpose
File Navigation	cd, ls, pwd	Navigate the file system
Permissions	chmod, chown	Manage file/directory access
Package Mgmt	apt, dpkg	Install/update software
Networking	ifconfig, ping, netstat, traceroute	Network diagnostics

5. Networking Basics

OSI Model: Studied all 7 layers: Physical, Data Link, Network, Transport, Session, Presentation, Application

TCP/IP Suite: Understood how TCP/IP protocols power the internet and local networks

DNS & HTTP/HTTPS: Learned how domain names resolve to IPs and how web traffic is encrypted

IP Addressing: Practiced subnetting, CIDR notation, and NAT concepts

6. Cryptography Basics

Topic	Description
Symmetric Encryption	Same key for encryption & decryption (e.g., AES)
Asymmetric Encryption	Public/Private key pair (e.g., RSA)
Hashing	MD5, SHA256 — one-way data transformation for integrity checking
SSL/TLS	Digital certificates securing HTTPS web traffic
Hands-on	Encrypted and decrypted messages using OpenSSL in Kali Linux terminal

7. Tool Familiarization

Tool	Purpose	Status
Wireshark	Packet capture and network traffic analysis	✓ Explored
Nmap	Network scanning — port and service detection	✓ Used
Burp Suite	Web proxy for intercepting HTTP/HTTPS traffic	✓ Familiarized
Netcat	Network debugging and connection testing	✓ Familiarized

8. Nmap Scan Results on Metasploitable2

An Nmap service version scan (-sV) was performed on Metasploitable2 from Kali Linux. The following open ports and vulnerable services were discovered:

Port	Protocol	Service	Version	Risk Level
21	TCP	FTP	vsftpd 2.3.4	CRITICAL
22	TCP	SSH	OpenSSH 4.7p1	Medium
23	TCP	Telnet	Linux telnetd	High
80	TCP	HTTP	Apache 2.2.8	High
139/445	TCP	SMB	Samba 3.x	Critical
3306	TCP	MySQL	MySQL 5.0.51a	High
5432	TCP	PostgreSQL	PostgreSQL 8.3	Medium
8180	TCP	HTTP	Apache Tomcat	High

9. Deliverables Completed

Deliverable	Status	Details
Lab Setup Report	✓ Done	This document with screenshots
GitHub Repository	✓ Done	Notes & Linux cheat-sheet uploaded
5-min Video Walkthrough	✓ Done	Screen recording of lab setup
Wireshark Test Capture	✓ Done	Captured network packets
Nmap Scan	✓ Done	Scanned Metasploitable2

10. Conclusion

Task 1 has been successfully completed. A fully functional cybersecurity lab has been set up with Kali Linux as the attacker machine and Metasploitable2 as the target, both running on VirtualBox with a Host-Only network configuration. Core concepts of cybersecurity (CIA Triad, threat types, attack vectors), Linux fundamentals, networking, cryptography, and essential tools (Wireshark, Nmap, Burp Suite, Netcat) have been studied and practiced. The lab is now ready for Task 2: Network Security and Scanning.